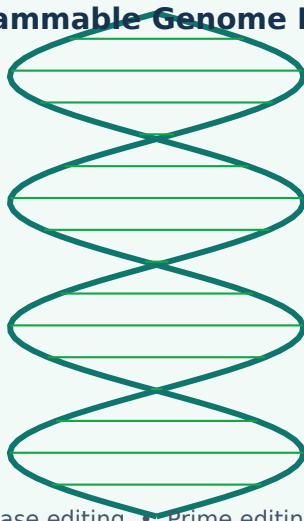


Recent Advancements in Genetic Engineering

CRISPR/Cas9 • Base Editing • Prime Editing • Gene Therapy

Programmable Genome Editing



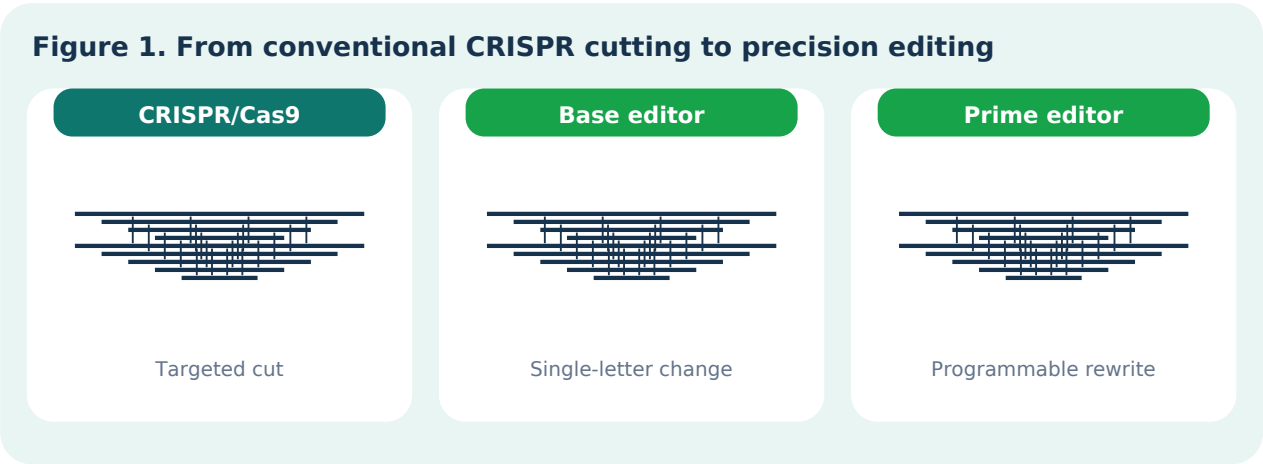
CRISPR/Cas9 • Base editing • Prime editing • Gene therapy

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Report Scope	Recent scientific and clinical developments
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Executive Snapshot

Clinical milestone	Casgevy became the first FDA-approved CRISPR/Cas9 therapy and later received an expanded pediatric indication.
Precision tools	Base editors can change selected DNA bases without a conventional double-strand break; prime editors can install substitutions and small insertions/deletions.
Delivery frontier	In vivo editing is moving gene editing from edited cells outside the body toward direct delivery to organs and tissues.
Computational acceleration	AI and machine learning are increasingly used for guide design, editor engineering and discovery of new CRISPR systems.
Main barriers	Off-target effects, delivery, immune responses, manufacturing, long-term safety, cost and equitable access remain central challenges.



The modern CRISPR toolkit is expanding from programmable DNA cutting toward base conversion, sequence rewriting and broader regulatory control.

Abstract

Genetic engineering has moved from a mainly experimental discipline to a clinically relevant biotechnology platform. The most important recent change has been the development of programmable CRISPR-based editors that can modify DNA with increasing precision and the translation of these technologies into human therapies. This report reviews recent advances in CRISPR/Cas9, base editing, prime editing, in vivo editing, delivery systems, artificial intelligence-assisted editor design, and gene therapy. A major milestone was the regulatory approval of Casgevy, the first CRISPR/Cas9-based therapy, for sickle cell disease and transfusion-dependent beta-thalassemia. In 2026, the FDA expanded Casgevy's approval to patients as young as two years old. At the same time, in vivo base editing has entered human clinical development, including approaches targeting PCSK9 for inherited hypercholesterolemia. Prime editing continues to evolve as a method for making substitutions, insertions and deletions without conventional double-strand DNA breaks. These advances are supported by improved delivery technologies and computational methods, including machine learning for guide RNA selection and editor engineering. Despite the promise, important challenges remain, including off-target editing, delivery to specific tissues, immune responses, manufacturing complexity, long-term safety, cost and equitable access. Overall, genetic engineering is entering a precision era in which the goal is shifting from simply cutting DNA toward programmable correction, regulation and rewriting of genetic information.

1. Introduction

Genetic engineering refers to the deliberate modification of an organism's genetic material to alter a biological function or correct a disease-causing change. Modern genome editing addresses earlier limitations by using programmable molecular systems that recognize a chosen DNA sequence. CRISPR-associated technologies have become central to this transformation because they are comparatively adaptable, efficient and programmable.

CRISPR systems were originally characterized as part of bacterial defense against invading genetic material. Their biotechnology potential became clear when researchers developed RNA-guided Cas9 as a programmable DNA nuclease. Since then, the field has expanded beyond conventional DNA cutting. Current platforms include nuclease-based editing, base editing, prime editing, epigenome editing, RNA editing and systems capable of inserting larger DNA sequences. Recent reviews emphasize that the modern CRISPR toolkit now spans genome, epigenome and transcriptome engineering (Villiger et al., 2024).

2. CRISPR/Cas9 and the Transition to Clinical Gene Therapy

Conventional CRISPR/Cas9 editing uses a guide RNA to direct Cas9 to a complementary DNA sequence next to an appropriate protospacer adjacent motif. Cas9 creates a targeted double-strand break. Cellular repair pathways then determine the final outcome. Non-homologous end joining can disrupt a gene through small insertions or deletions, while homology-directed repair can support more precise sequence changes when a suitable template is available.

The most important clinical milestone is exagamglogene autotemcel, marketed as Casgevy. In December 2023, the U.S. Food and Drug Administration approved Casgevy for patients aged 12 years and older with sickle cell disease and recurrent vaso-occlusive crises. It was the first FDA-approved treatment using CRISPR/Cas9 genome editing (U.S. Food and Drug Administration [FDA], 2023). In January 2024, the indication was expanded to transfusion-dependent beta-thalassemia. The treatment uses the patient's own hematopoietic stem cells. These cells are edited outside the body to alter a regulatory region associated with BCL11A, a regulator of fetal hemoglobin. After conditioning, the edited cells are infused back into the patient and engraft in the bone marrow. Increased fetal hemoglobin can reduce the tendency of red blood cells to sickle and can improve anemia in beta-thalassemia (FDA, 2024).

In July 2026, the FDA expanded Casgevy's approval to patients aged two years and older for both sickle cell disease with recurrent vaso-occlusive crises and transfusion-dependent beta-thalassemia (FDA, 2026). This demonstrates the continuing movement of genome editing from proof-of-concept toward broader clinical use. However, the treatment remains medically complex because patients require intensive conditioning and careful monitoring. FDA prescribing information also includes warnings related to hematologic complications and the possibility of unintended genome editing.

Clinical significance: Casgevy shows that CRISPR genome editing has crossed from experimental research into regulated human therapy, while the 2026 pediatric expansion illustrates continued clinical translation.

3. Base Editing: Changing Individual DNA Letters

Base editing is one of the most important developments beyond nuclease-based CRISPR. Instead of producing a conventional double-strand break, a base editor combines a modified Cas protein with a deaminase enzyme. Depending on the system, it can convert one DNA base into another, such as C to T or A to G, within an appropriate editing window. This approach is particularly attractive for diseases caused by single-nucleotide variants.

The advantage of base editing is that many desired changes can be made without generating a double-strand DNA break or requiring a donor DNA template. This may reduce some forms of unwanted DNA damage. Nevertheless, base editors have limitations. Their editing window and target-sequence requirements can restrict which mutations are accessible. They may also produce bystander edits near the target and, in some designs, unintended changes outside the intended site. Therefore, specificity assessment remains essential (Villiger et al., 2024).

Recent clinical development has included base-editing approaches for blood disorders and inherited lipid disorders. One notable direction is in vivo editing of PCSK9 in the liver. PCSK9 regulates the abundance of LDL receptors, so reducing PCSK9 activity can lower LDL cholesterol. In 2026, a clinical report described in vivo base editing with VERVE-102 for hypercholesterolemia, demonstrating the growing feasibility of delivering a gene editor directly into the body rather than removing and editing cells outside the patient (Vafai et al., 2026).

4. Prime Editing: Programmable Genetic Rewriting

Prime editing was introduced as a more versatile form of precision genome editing. A prime editor combines a Cas9 nickase with a reverse transcriptase and uses a specialized prime-editing guide RNA. Rather than creating a conventional double-strand break, the system can write a new DNA sequence into the target region. Prime editing can support targeted base substitutions as well as small insertions and deletions, making it potentially useful for a broad range of pathogenic variants (Murray et al., 2025).

Recent work has focused on improving editing efficiency, reducing unwanted by-products, expanding the range of targetable sequences and developing better delivery systems. Reviews published in 2025 and 2026 describe an increasingly sophisticated family of prime editors, including engineered variants and improved guide-RNA designs (Lee et al., 2025; Lin, 2026).

Prime editing remains mainly a research and translational technology rather than an established routine therapy. Its large molecular components create delivery challenges, and editing efficiency can vary considerably between cell types and genomic sites. Nevertheless, its ability to make multiple classes of precise changes without a conventional double-strand break makes it an important candidate for future genetic medicines.

Precision principle: conventional CRISPR often relies on a DNA break followed by cellular repair, whereas prime editing is designed to write programmed sequence changes without a conventional double-strand break.

5. In Vivo Editing and Delivery Technologies

A major frontier in genetic engineering is the shift from ex vivo editing to in vivo editing. Ex vivo approaches, such as Casgevy, allow cells to be collected, edited and tested before being returned to the patient. This provides substantial control but is labor-intensive and difficult to apply to tissues whose cells cannot easily be removed and reinfused.

In vivo editing aims to deliver the editor directly to the relevant organ or tissue. Lipid nanoparticles, viral vectors and other delivery platforms are being investigated. The choice of delivery system affects tissue targeting, dose, duration of expression and safety. The NIH's 2026 TARGETED Challenge highlights delivery as a major remaining bottleneck. The initiative focuses on programmable delivery and non-viral delivery across the blood-brain barrier, reflecting the fact that a powerful editor is not clinically useful if it cannot reach the correct cells in sufficient quantity (National Institutes of Health [NIH], 2026).

6. Artificial Intelligence and Next-Generation Editors

Artificial intelligence is becoming an important supporting technology in genetic engineering. Machine learning models can help predict guide RNA activity, identify potential off-target sites, analyze sequence context and prioritize promising editor designs. AI can also support protein engineering by predicting how sequence changes may alter the properties of CRISPR-associated proteins.

A 2026 Nature Reviews Genetics review described the growing integration of AI with CRISPR technology, including guide design, engineering of existing editors and discovery of new genome-editing enzymes. The combination of large biological datasets with machine learning may make editor development faster and more systematic. In the longer term, AI-based models may help predict how a particular genetic edit will affect cell behavior (Thomson et al., 2026).

AI does not remove the need for laboratory validation. Predictions depend on the quality and diversity of training data, and biological systems can behave differently across cell types and individuals. Experimental

confirmation of on-target activity, off-target effects and functional outcomes remains essential.

7. Applications Beyond Human Medicine

Genetic engineering is also transforming agriculture, biotechnology research and industrial applications. CRISPR can be used to modify plants for disease resistance, stress tolerance, nutritional traits and improved productivity. In research laboratories, genome editing helps scientists investigate gene function by selectively disrupting, activating or modifying genes. These applications accelerate the study of disease mechanisms and the development of biological models.

In industrial biotechnology, precise genetic engineering can optimize microorganisms for the production of enzymes, metabolites, biofuels and other valuable compounds. The ability to edit metabolic pathways may improve yields and reduce unwanted by-products. These applications demonstrate that genome editing is not limited to therapeutic medicine. It is becoming a general platform for biological design.

8. Challenges and Ethical Considerations

The rapid development of genetic engineering also creates scientific, ethical and social questions. Off-target editing is a central safety concern because unintended genomic changes could have harmful consequences. Even an on-target edit can sometimes produce unexpected deletions, rearrangements or other genomic changes. Modern safety assessment therefore requires more than checking whether the intended sequence was changed.

Delivery and immune responses are additional challenges. A therapeutic editor must reach the correct cells while limiting exposure elsewhere. Some components of editing systems may also interact with the immune system. Manufacturing is another major issue because personalized cell therapies require sophisticated facilities, quality control and specialized clinical infrastructure.

Cost and equitable access are equally important. A one-time genetic medicine may have a high upfront cost even when it is designed to provide a long-term benefit. Countries with limited health-care resources may face difficulties establishing the infrastructure needed for complex cell and gene therapies. Ethical governance must therefore consider not only whether an intervention is scientifically possible, but also who can safely access it.

Germline editing raises an additional ethical boundary because changes made to reproductive cells or embryos could be inherited by future generations. Most clinical applications discussed today focus on somatic cells, where the intended genetic change is limited to the treated individual. Continued international discussion and responsible regulation are essential as editing technologies become more powerful.

9. Future Prospects

The next phase of genetic engineering is likely to emphasize precision, tissue-specific delivery, lower toxicity and more predictable outcomes. Base and prime editors may expand the range of mutations that can be corrected without conventional DNA double-strand breaks. Improved delivery systems could make in vivo editing practical for more organs. Compact editors and novel CRISPR-associated proteins may also expand the number of genomic targets that can be reached.

Another important direction is the integration of genome editing with computational biology, single-cell analysis and systems biology. Instead of treating a gene as an isolated target, future therapies may use broader information about cell state, regulatory networks and disease mechanisms.

10. Conclusion

Recent advancements in genetic engineering show a clear transition from basic genome cutting toward increasingly programmable and clinically useful genetic modification. CRISPR/Cas9 has already crossed a major regulatory boundary through Casgevy, while base editing and prime editing are expanding the types of changes that can be made. In vivo editing and improved delivery systems are addressing the challenge

of reaching tissues directly, and artificial intelligence is helping researchers design and optimize editors.

At the same time, genome editing should not be viewed as a risk-free technology. Off-target effects, delivery limitations, immune responses, manufacturing complexity, cost and ethical concerns remain significant. For biotechnology students and researchers, this field illustrates how molecular biology, bioinformatics, medicine and engineering are increasingly converging to create new approaches for understanding and treating disease.

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Source note: The report prioritizes regulatory information from the U.S. FDA, current scientific reviews, and a 2026 clinical report. Scientific claims are summarized in original wording and supported with APA-style in-text citations.